

Towards a Strategic Theory of the Firm

This is the original LaTeX version of the paper prepared for the 1981 conference “Non-Traditional Approaches to Policy Research,” held at the USC School of Business on November 12-13, 1981.

It and most of the conference papers were later published as Rumelt, R. P. 1984. “Towards a Strategic Theory of the Firm.” In: Robert Lamb (ed.) *Competitive Strategic Management*. Ed R. Lamb. Englewood Cliffs, NJ: Prentice-Hall, 556–70. Unfortunately, the typesetter for that volume had no skill at mathematics, and Figure 1 was clobbered.

A cleaner version was reprinted in Foss, Nicolai J., ed. *Resources, firms, and strategies: a reader in the resource-based perspective*. Oxford University Press, 1997, 131-45.

The published versions cite “Uncertain Imitability” in the Bell Journal of Economics, but in the original working paper, the reference was to the unpublished working paper of the same title. Also, the original working paper references “How Important is Industry in Explaining Firm Profitability.” That was a 1975 working paper on firms rather than on lines of business. It was rejected by the AER and is now lost. Later versions should have referenced Rumelt, Richard P. “How Much Does Industry Matter?” *Strategic Management Journal*, 1991, but were somehow not updated.

Towards a Strategic Theory of the Firm

Version 3

Richard P. Rumelt

October 7, 1981

I consider myself a mainstream researcher in the field of business policy, and the ideas I want to describe in this paper concern the foundations of a theory of business strategy that is rooted in economics. But is such a paper, whatever its merits, really appropriate at a conference entitled "Non-traditional Approaches to Policy Research"? Surprisingly, it is. The use of economic theory to model and explicate business strategy, as it is understood within the field of business policy, is distinctly non-traditional.

To the uninitiated, it appears obvious that the study of business strategy must rest on the bedrock foundations of the economist's model of the firm and the theory of industrial organization. Nevertheless, until very recently, there has been a virtually complete absence of any intersection between business policy and economic theory. Few attempted a cross-fertilization between these fields; most who did reported results barren of much interest. This state of affairs did not reflect the unwillingness of policy researchers to learn economics. Rather, it came about because the neoclassical theory of the firm was created by assuming away the very existence of those phenomena that most concern students of business policy. It was sustained by the mainstream microeconomist's commitment to the position that the theory of the firm not be required to describe the actual behavior of firms.

This situation is beginning to change. The pioneering insights of Coase, Simon, and Stigler set in motion forces that are undermining the neoclassical theory.¹ Recent work by Williamson, Porter, Spence, and others demonstrates that economic concepts can model and describe strategic phenomena.² In this paper I want to look closely at this new confluence and suggest the outlines of a "strategic" theory of the firm.

¹Ronald Coase, "The Nature of the Firm," *Economica*, 4 (November 1937), 386-405; H. A. Simon, *Administrative Behavior*, 2nd ed. (New York: Macmillan, 1961); and George J. Stigler, "The Economics of Information," *Journal of Political Economy*, 69 (June 1961), pp. 213-25.

²Oliver E. Williamson, *Markets and Hierarchies: Analysis and Antitrust Implications* (New York: Free Press, 1975); Michael E. Porter, *Competitive Strategy* (New York: Free Press, 1980); and A. M. Spence, "Investment, Strategy, and Growth in a New Market," *Bell Journal of Economics*, 10 (Spring 1979).

Business Policy and Neoclassical Theory

Business policy is concerned with those aspects of general management that have material effects on the survival and success of business enterprises. For about twenty years the central organizing concept in this field has been strategy. The concept is empirical rather than theoretical, having its roots in numerous field studies of business firms and the historical analysis of the evolution of business enterprises. In essence, the concept is that a firm's competitive position is defined by a bundle of unique resources and relationships and that the task of general management is to adjust and renew these resources and relationships as time, competition, and change erode their value. This way of looking at the firm is not a theory; it is a set of constructs that have proved useful in describing and summarizing the empirical studies of firm behavior that form the core of the business policy literature. Simply put, these broad empirical observations are:

1. The general managers of firms make choices, and some of these choices are considerably more important (having more impact on performance) than others.
2. Strategic choices are not necessarily explicit but may be characterized by infrequency, uncertainty, the irreversibility of commitments, and multi-functional scope, and they are usually nonrecurring.
3. The most critical strategic choices exhibited by a firm are those concerned with the selection of the product-market areas or segments in which the firm will compete and the basic approach to those businesses.
4. Similar firms facing similar strategic problems may respond differently.
5. Firms in the same industry compete with substantially different bundles of resources using disparate approaches. These firms differ because of differing histories of strategic choice and performance and because managements appear to seek asymmetric competitive positions.

What makes these apparently innocuous observations worth repeating is that they either contradict or stand completely outside the neoclassical theory of the firm and the standard models used in industrial organization.³ For example, Cohen and Cyert's treatment of the theory of the firm begins this way:

This book concentrates on the most significant institution in our economic system—the business firm—and the most important function of our economic system—resource allocation. It is impossible to

³They also contradict the environmental determinism viewpoint in organizational sociology. It is an intriguing example of disciplinary chauvinism that organization sociologists tend to attribute the straightforward notion that senior management's choices influence the selection of an organization's tasks and internal structure to Child's strong reminder on this point. See John Child, "Organization Structure, Environment and Performance—The Role of Strategic Choice," *Sociology*, 6 (January 1972), pp. 1–22.

have a clear understanding of the functioning of the modern American society without knowing a great deal about the firm and the role it plays in the economy's resource allocation process.⁴

Nevertheless, in the following four hundred pages of this well-written text, few real firms are mentioned and no model is discussed in which the firm's choice problem is anything other than the selection of a price or level of output for a homogeneous product with known production and demand functions. Monopolistic competition is discussed but rejected on theoretical and empirical grounds.

The problem with the neoclassical theory is that it is not really a theory of the firm. The existence of the firm is actually problematic within the axiomatic framework of the theory and must be justified by reference to entrepreneurship as a fixed factor. What the theory actually deals with is the workings of the price system in a setting in which nothing but prices need be known. It is a powerful intellectual achievement, but that power was obtained by assuming away such phenomena as (1) transaction costs, (2) limits on rationality, (3) technological uncertainty, (4) constraints on factor mobility, (5) limits on information availability, (6) markets in which price conveys quality information, (7) consumer or producer learning, and (8) dishonest and/or foolish behavior.

In fact, many of the assumptions embodied in the neoclassical theory are only now coming to light as specialized models of markets are being investigated. One interesting example is Spence's discussion of signaling equilibria,⁵ in which it is sensible to invest in otherwise useless activities in order to signal quality differentials. Another is Diamond's observation that the introduction of a small consumer search cost into a retail market model drastically alters the nature of the equilibrium.⁶

The situation with regard to industrial organization has been only marginally better. Within industrial organization there is a subschool which, like business policy, has recorded and commented on a wide variety of real-world business behavior. However, the theoretical structure of the field has never encompassed that richness. The traditional model of industry in industrial organization is taken from oligopoly theory⁷ and remains that of identical firms or firms that are homogeneous but for scale. The effect of this modeling assumption has been to reduce the study of industrial competition to the study of relative scale, all other differences being ignored.

Industries differ, in the standard view, according to the degree of concentration, the presence of scale economies, and the degree of product differentia-

⁴Kalman J. Cohen and Richard M. Cyert, *Theory of the Firm: Resource Allocation in a Market Economy* (Englewood Cliffs, N.J.: Prentice-Hall, 1965), p. 3.

⁵A. M. Spence, *Market Signaling: Informational Transfer in Hiring and Related Processes* (Cambridge, Mass.: Harvard University Press, 1974).

⁶Peter A. Diamond, "A Model of Price Adjustment," *Journal of Economic Theory*, 3 (1971), 156-68.

⁷In this regard it is important to note that oligopoly theory, far from providing a predictive model of firm behavior, represents the "edge" of economics in that a profit-maximizing assumption fails to define behavior. Oligopoly models are constructed by first assuming a pattern of behavior and then deducing the form of the resultant equilibrium, if one exists.

tion or nonprice competition. Scale economies and product differentiation are viewed, in this framework, as properties of an industry rather than as the results achieved by firms and both are identified as barriers to entry, creating economic rents (profits that do not attract new production).

By taking the industry as the unit of analysis, industrial organization has largely ignored the theory and evidence of intra-industry differences among firms. Thus, while great efforts have gone toward explaining interindustry differences in the rate of return, it can be easily shown that the dispersion in the characteristic long-term rates of return of firms within industries is five to eight times as large as the variance in returns across industries.⁸

What has caused this mismatch between the concept of strategy and neoclassical theory? To say that the economic models are not "realistic" would be true, but unfair and beside the point. All models in the social sciences are unrealistic; good models are specialized and simple, generating descriptions of the phenomena of interest with the fewest necessary constructs and assumptions. This mismatch arises because policy researchers and economists have been interested in substantially different phenomena. The central concerns of business policy are the observed heterogeneity of firms and the firm's choice of product-market commitments. By contrast, the basic phenomena of interest in neoclassical theory is the functioning of the price system under norms of decentralized decision making.

In the language of economics, the chief concern of business policy researchers has not been static profit maximization⁹ but profit seeking through corporate entrepreneurship and with the empirical observation that corporate entrepreneurship is intimately connected with the appearance and adjustment of unique and idiosyncratic resources. This view of corporate behavior is most closely associated with Schumpeter's vision of competition as the process of "creative destruction" rather than as a static equilibrium condition.¹⁰ As Nelson and Winter have noted:¹¹

The core ideas of Schumpeterian theory are of course quite different from those of neoclassical theory. For Schumpeter the most important firms are those that serve as the vehicles for action of the real drivers of the system—the

⁸See R. P. Rumelt, "How Important Is Industry in Explaining Firm Profitability?" U.C.L.A., 1981.

⁹The standard attack on the neoclassical theory has centered on the assumption of profit maximization. According to the behavioral theory of the firm explicated by Cyert and March, managers are only intendedly rational. Bounds on their information-processing capacities prevent firms from acting as true profit maximizers. Baumol's somewhat different argument is that the quantity corporations act to maximize is not really profits but sales (or growth, or some measure of manager's welfare). Current fashion terms these adjustments to the neoclassical model as highlighting information costs, computation costs, and agency problems. See Richard M. Cyert and James G. March, *A Behavioral Theory of the Firm* (Englewood Cliffs, N.J.: Prentice-Hall, 1963); and William J. Baumol, *Business Behavior, Value and Growth* (New York: Macmillan, 1959).

¹⁰J. A. Schumpeter, *The Theory of Economic Development* (Cambridge, Mass.: Harvard University Press, 1934; original publication 1911).

¹¹R. R. Nelson and S. G. Winter, "Neoclassical vs. Evolutionary Theories of Economic Growth: Critique and Prospectus," *Economic Journal*, 84 (December 1974), 890.

innovating entrepreneurs ... The competitive environment within which firms operate is one of struggle and motion. It is a dynamic selection environment, not an equilibrium one. The essential forces of growth are innovation and selection, with augmentation of capital stocks more or less tied to these processes.

Thus the concepts of entrepreneurship and resource heterogeneity, so central to business strategy, have been either omitted or taken as preexisting givens

in neoclassical theory. Entrepreneurship is normally omitted because it has become by definition the repository of nonneoclassical phenomena. Entrepreneurs are seen to possess special information, to be unique, to create pure profit, and to act as the essential indivisibilities governing the size distribution of firms. Similarly, resource heterogeneity is taken to be an exogenous property of the physical world rather than an endogenous creation of economic actors. Their omission has obscured the close logical connections between these constructs. Without resource heterogeneity (and the equivalent of property rights to unique resources), there is little incentive for investing in the risky exploration of new methods and the search for new value. Given uncertainty, the ex post results of entrepreneurial activity will necessarily be resource heterogeneity. At the most primitive level, firms may simply differ in the relative efficiency with which they extract or process homogeneous goods. However, in the absence of perfect intermediate markets for these goods, firms will have incentives to integrate. Thus is born the strategic firm, characterized by a bundle of linked and idiosyncratic resources and resource conversion activities.

What accounts for the neglect of these important phenomena? Nordhaus and Tobin have argued that the problem is one of modeling:¹²

Many economists agree with the broad outlines of Schumpeter's vision of capitalist development, which is a far cry from growth models made nowadays in either Cambridge, Mass. or Cambridge, England. But visions of that kind have yet to be transformed into theory that can be applied to everyday analytic and empirical work.

I agree that there are substantial difficulties in creating formal models of Schumpeterian rivalry, but I do not see them as insurmountable. There may be, however, another factor deterring economists from such efforts—determining the welfare implications of Schumpeterian rivalry is markedly more difficult than modeling it. This implies that policy researchers, who are only secondarily concerned with the technical welfare implications of competitive activity, may well have a comparative advantage at producing descriptive theory in this area.

¹²W. Nordhaus and J. Tobin, "Is Growth Obsolete?" in *Economic Research: Retrospect and Prospect, Economic Growth*, ed. R. Gordon (New York: National Bureau of Economic Research, 1972), p. 2. Also quoted in Nelson and Winter, "Neoclassical vs. Evolutionary Theories," p. 889.

Uncertain Imitability and Ambiguity

This section describes a simple theory of rivalry under conditions of causal ambiguity. I and my colleague Steven Lippman have named the theory and modeling concept it embodies "uncertain imitability." We model entrepreneurship as

the production of new production functions and generate firm heterogeneity as an outcome rather than as a given. The treatment here is only a rough sketch of the more complete theory, and interested readers are directed to the source.¹³

In neoclassical theory firms entering industries or undertaking substantial expansion efforts select their production functions from a known bundle of technological possibilities. Thus imitative attempts tend to equilibrate firm efficiencies, and long-term differences in profitability signal failures in either product or factor markets. But suppose that there exists an irreducible uncertainty connected with the creation (or production) of a new production function. Then the efficiencies achieved by entrants or major expansion programs will vary. Furthermore, if there is a nonrecoverable cost associated with such entrepreneurial activities, rational actors will stop short of seeking to imitate the best extant firm. Thus the ambiguity that generates the initial heterogeneity will also act to block its homogenization through imitation.

Uncertainty in the creation of new production functions is most likely to come about because there is ambiguity as to what the factors of production actually are and as to how they interact. The standard neoclassical assumption is that there is a finite set of known factors of production and that their marginal productivities can be discerned. However, if the precise reasons for success or failure cannot be determined, even after the event has occurred, there is causal ambiguity and it is impossible to produce an unambiguous list of the factors of production, much less measure their marginal contributions.

In order to model this situation, assume that a market exists for a homogeneous product. There is a known demand function and any entrant into the industry obtains a cost function $C(q, b)$ which is U shaped in q , the rate of output. The parameter b is a measure of relative efficiency and larger values of b imply lower unit costs. Each firm, once it is in the industry, acts as a profit-maximizing price taker. By adjusting output so that marginal cost equals the market price p , a firm with efficiency b obtains a level of profits $y(b, p)$.

If $C(q, b)$ exhibits fixed costs, firms may display negative profits when the market price falls below some level. Define $h(p)$ as the value of b that just allows the firm to break even at a market price p . Thus $y(h(p), p) = 0$, and if $b < h(p)$, the firm cannot make positive profits and is forced to withdraw from the industry. Now assume that each entrant into the industry must pay a nonrecoverable "entry fee" K and then receives a cost function $C(q, b)$ in which b is a realization of a random variable X with cumulative distribution F and density f . Intuitively, it should be clear that firms will enter the industry and display a variety of efficiencies. As entry continues, the price will fall,

¹³S. A. Lippman and R. P. Rumelt, "Uncertain Imitability: An Analysis of Interfirm Differences in Efficiency under Competition," UCLA GSM Working Paper, 1981.

forcing some firms that have received "poor" cost functions (low b 's) out of the industry. Finally, the industry will reach a state that deters further rational entry attempts. In a sense, the final free-entry

equilibrium is achieved through processes of variation and selection rather than neoclassical resource flows. In it, firms display a range of efficiencies and the most efficient display stable persistent rents.

In general it is quite difficult to characterize the final equilibrium in that the optimal entry strategy depends in detail upon the mix of extant firms at each point. For simplicity, I will assume that each entrant has a negligible independent impact on the market price. Now define $V(p)$ as the expected value of entry when the prospective entrant faces an unchanging price p . Taking the discount rate to be r , we have

$$V(p) = -K + \frac{1}{r} \int_{h(p)}^{\infty} y(x, p) dF(x) \quad (1)$$

For p sufficiently small, $V(p) < 0$ and entry is blocked—even the guarantee of a fixed price being insufficient to make entry attractive. Because V increases in p , there is a unique price p^* that solves $V(p) = 0$, and atomism¹⁴ ensures that the final entry-blocking equilibrium price will only be infinitesimally less than p^* . Each prospective entrant can therefore expect to receive at least $V(p^*)$ and entry will proceed as long as $p > p^*$.

When the price is p^* , the minimum b necessary for survival is $b^* = h(p^*)$. Firms in the final equilibrium will be those that received values of b greater than b^* . Thus the probability P_s that an entrant will survive the selection process and be a member of the final equilibrium is

$$P_s = 1 - F(b^*) \quad (2)$$

The distribution F_s of survivors' b 's is $F(x)/P_s$ for $x > b^*$ and zero otherwise. Survivors of the selection process earn profits $y_s(b, p^*)$ in perpetuity and the expected level of survivors' profit is

$$Ey_s = \frac{1}{P_s} \int_{b^*}^{\infty} y(x, p^*) dF(x) \quad (3)$$

Combining (1) with (3) and the condition $V(p^*) = 0$ gives

$$Ey_s = rK/P_s \quad (4)$$

and defining survivors' rents as $R_s = Ey_s - rK$ reveals that

$$R_s = rK(1 - P_s)/P_s \quad (5)$$

¹⁴The nonatomistic free-entry equilibrium can be characterized if firms do not drop out of the industry. This result is obtained if fixed costs are zero, sunk, or scaled with the rest of the firm. With this specification, entry produces a renewal process in total industry capacity. For details, see Lippman and Rumelt, "Uncertain Imitability."

From (5) it is clear that the average survivor (and the industry as a whole) displays positive economic profits in equilibrium despite price-taking behavior and free entry. These rents are created by the failure of other firms to survive: if $P_s = 1$, then $R_s = 0$. More precisely, for each survivor there are $(1 - P_s)/P_s$ firms that fail to survive, each suffering a loss K . Consequently the net expected profit from an entry attempt is zero, and the survivors' rents exactly balance the failures' losses.

Another way of looking at (5) is that selection pressure ensures that the "average" entrepreneur will earn below-average profits. Indeed, if P_s is low enough, the mean of X will fall below b^* and the average entry attempt will not only earn poor profits but fail to survive.

In this equilibrium the extant firms have been selected for their unusual efficiency at production, and no outsider can expect to beat them at their own game. This result does not depend upon scale economies or sunk costs. Rather, it is the natural consequence of a nonneoclassical selection-based equilibrium. In the language of business policy, the more-efficient firms have "created" or discovered unique skills and strengths and will maximize their values by seeking other areas of activity wherein these special skills may also be of value.

What are the implications of increased riskiness? Figure 1 shows the shape of $y_s(b, p^*)$. Note that if y_s is not convex in b for $b > b^*$, it is always possible to find a monotone function $b \rightarrow b_1$ together with its associated distribution and density functions, such that $y_s(b_1, p)$ is convex. This means that it is the riskiness of y rather than b that carries economic meaning. For simplicity, assume that the parameterization of $C(q, b)$ is chosen so that $y_s(b, p^*)$ is linear in b for $b > b^*$.

Given y_s convex in b , it can be seen from (1) that mean-preserving increases in the spread of F (or increases in variance) will tend to decrease the equilibrium price p^* . Equivalently, were such a pure increase in riskiness available privately to a prospective entrant, it would increase the expected value of that entry attempt. This "risk-loving" behavior also occurs in the case of call options; ceteris paribus, the owner of a call option prefers mean-preserving increases in risk to decreases. The principle also applies to the levered firm, a mean-preserving increase in risk shifting wealth from bondholders to stockholders.

A Theory of Firm Size

The relationship between riskiness and private value has interesting implications for the optimal scope of entrepreneurial firms or risky projects. To begin, notice that irreversibly combining two separate entry attempts (paying 2 K and receiving

costs $C(b_1, q_1) + C(b_2, q_2)$ is never attractive. Intuitively, the relative draws from F are independent so that the relative riskiness has decreased, making the expected value of entry negative. Irreversibly combining two entry attempts would only be efficient if the draws from F could be guaranteed to be completely correlated. These considerations suggest that the entrepreneur's problem is not only to decide whether or not to enter but also to assemble the appropriate mix

of assets for the attempt.

Which activities should the entrepreneur combine? The general answer is, Those that will exhibit strongly dependent postentry efficiencies. Given a bundle of activities with total postentry efficiency determined by the random variable X , adding new activities that involve sunk costs can never be profitable if their efficiencies are uncorrelated with X . Thus new activities are added until the point where further additions would not add sufficiently large expected profits or profit variance to justify the added sunk capital.

Note that this logic is opposite to that underlying diversification. Here the entrepreneur's objective is not to reduce risk, as the expected level of efficiency provides an inadequate return. Rather, the entrepreneur is attempting to concentrate only on those activities that are closely connected with success or failure in the market. Adding extraneous activities not only produces an unnecessary increase in the costs of failure but also may obscure important information regarding the success of the main endeavor.

Interestingly, this perspective provides a theory of firm size that does not depend upon diseconomies of scale or control loss and is only tangentially related to the notion of a fixed entrepreneurial factor. In addition, it explains why diversification, which reduces the risk of bankruptcy, is rarely undertaken by those facing the greatest risk—entrepreneurs entering or creating new markets.

Selection, Adaptation, and Isolating Mechanisms

Given uncertain imitability, in equilibrium the average firm earns positive economic profit and is more efficient at what it is doing than a new entrant should expect to be. Were this not true, entry would proceed until it was true. If the equilibrium is permeated by heterogeneous firms with evolved local advantages, there are a number of immediate implications. Among the most straightforward are these:

1. New entry activity will essentially be a function of market growth rather than industry profitability. High levels of profitability in stable markets may well signal incumbents possessing difficult-to-imitate skills and deter entry.
2. Firms that are successful in one endeavor will tend to seek out related activities in which their revealed special competences are useful.¹⁵ Hence profitability and growth will be correlated even when the effects of demand pull are controlled.
3. If the basis for success in a market shifts to a new function, firms that have been successful in the past may now be at a disadvantage relative to

¹⁵Strictly speaking, a resource or competence will be fully utilized in one sphere of activity if it is a factor of production for a perfect commodity. However, if firm size is limited by differentiation, segmentation, and other exogenous sources of declining marginal revenue, unique factors may not be fully utilized in a single activity.

out side firms possessing demonstrated skills related to the new required competence.

In industrial organization theory a barrier to entry exists when a prospective entrant is at a disadvantage relative to an incumbent.¹⁶ In an important paper, Caves and Porter extend the entry barrier concept by defining mobility bar-

riers as asymmetries among firms within industries that act to limit differential expansion and the equalization of profit rates. Their emphasis on heterogeneity deserves special attention:

Limits on entry and limits on mobility remain stubbornly immiscible as long as we stick to conventional thinking about cost functions and intra-industry differences among firms. The conventional approach takes firms within an industry as identical in all economically important respects except for their size. Then cost conditions either define an optimal scale for the firm, leaving no explanation why size should change, or render scale indeterminate, providing no clue as to what could deter infinitesimal changes in a firm's scale.... The key to conjoining barriers to entry to a more general theory of interscale mobility of firms is the hypothesis that sellers within an industry are likely to differ systematically in traits other than size."

In Caves and Porter's view, industries can be broken into groups of firms that exhibit distinct characteristics (e.g., broad-line vs. specialist). Mobility barriers both define these groups and are reinforced by the strategic activities of group members. The group concept is frequently all that is needed, but there is no theoretical reason to limit mobility barriers to groups of firms. I shall therefore use the term isolating mechanism to refer to phenomena that limit the ex post equilibration of rents among individual firms.

In the pure theory of uncertain imitability, the isolating mechanism is causal ambiguity. The inability of economic agents to fully understand the causes of efficiency differences limits competition by entry or imitation. However, many other isolating mechanisms exist. For example, mineral rights laws convert the results of risky exploration investments into ex ante uncertain but ex post persistent streams of rent. Similarly, patents, trademarks, reputation, and brand image serve to limit second-mover imitation of first-mover success. These and other important isolating mechanisms are listed in Table 1.

The importance of isolating mechanisms in business strategy is that they are the phenomena that make competitive positions stable and defensible. Many of them appear as first-mover advantages. For example, the first firm to commit idiosyncratic capital to serving a small market segment may gain a stable position serving that segment. From an equilibrium perspective, such events are inherently uncertain. That is, the existence of the market may have been

¹⁶Defining an entry barrier precisely is no simple task. Perhaps the most straightforward approach is to define it by its results—persistent surplus profits that do not induce entry. Stigler suggests that entry barriers exist when entrants would bear costs not borne by incumbents. This definition neatly separates the impact of scale economies, which are available to any producer, from sunk costs (idiosyncratic investment), which must enter into the entrant's prospective cost calculations but not those of the incumbent. See George J. Stigler, *The Organization of Industry* (Homewood, Ill.: Richard D. Irwin, 1968).

problematic, the technology may have been uncertain, the information necessary to make the early move may have been unpredictably distributed, or by first-mover we may simply mean the first successful mover.

Although isolating mechanisms provide (ex post) stable streams of rent,^{17,18}

Table 1: Elements of Strategic Position

Sources of Potential Rents	Isolating Mechanisms
Changes in technology	Causal ambiguity
Changes in relative prices	Specialized assets
Changes in consumer tastes	Switching and search costs
Changes in law, tax, and regulation; discoveries and inventions	Consumer and producer learning
	Team-embodied skills
	Unique resources
	Special information
	Patents and trademarks
	Reputation and image
	Legal restrictions on entry

the opportunities to create, "jump behind," or otherwise exploit them must arise from unexpected changes. Without uncertainty there is no wedge between the ex ante price of an asset or market position and its ex post value. It is the juxtaposition of isolating mechanisms with uncertainty that permits the modeling of heterogeneity in an equilibrium framework.

Table 1 emphasizes this point by also showing the important sources of potential rents. They are essentially unexpected changes in the environment. Thus Table 1 presents a simple theory of strategy: a firm's strategy may be explained in terms of the unexpected events that created (or will create) potential rents together with the isolating mechanisms that (will) act to preserve them. If either element of the explanation is missing, the analysis is inadequate.

Implications for Normative Theory

Are there normative implications of this view of strategy or does it place the fortunes of the firm in the hands of exogenous events and impersonal isolating mechanisms? My view is that there are very real normative implications and they can be based on much sounder theory than much of the currently popular prescription.

First, it should be clear that a firm's stability and profitability fundamentally depend upon entrepreneurial activity. There cannot be a simple algorithm for

¹⁷R. E. Caves and M. E. Porter, "From Entry Barriers to Mobility Barriers: Conjectural Decisions and Contrived Deterrence to New Competition," *Quarterly Journal of Economics*, May 1977, pp. 241-61.

¹⁸Ibid., p. 250.

creating wealth. Still, it is true that some ways of approaching the problem of strategy will be more fruitful than others. In particular, these points deserve emphasis:

1. The opportunities for strategic change occur infrequently, and their timing is largely beyond the control of management. The chance to substantially improve one's competitive position does not arise out of pricing or advertising tactics, but the recognition of change in some underlying factor.
2. Unexpected events may change the distribution of sales and profits within an industry, acting as windfall gains and losses to incumbents. It is vital that management recognize and take full advantage of these events. The routine component of strategy formulation is the constant search for ways in which the firm's unique resources can be redeployed in changing circumstances.
3. More fundamental shocks act to change the very structure of the industry, altering the nature and magnitudes of the isolating mechanisms at work. Examples of such events are airline deregulation, the advent of small computers, and the impact of oil prices on the world automobile market. In such situations it is usually unclear what the eventual structure of the industry will be. Firms that are lucky or insightful enough to make early commitments to what turn out to be defensible positions can be stunningly successful. The implication is that a developmental theory of industry structure would be of significant value.
4. A critical strategic question in a growth industry is the shape of the final equilibrium. When industry growth is rapid, profit rates are normally quite high, but reinvestment rates that are even higher work to produce net negative cash flows. If firms misjudge the strength of isolating mechanisms in the final equilibrium, the slowing of growth will bring profit rates to below-normal rates; the industry will have functioned as a cash trap. Theory and empirical work on this issue would have obvious normative value.
5. If opportunities for significant shifts in strategic position are infrequent, and if isolating mechanisms create defensible positions, it follows that many firms can ignore strategy for long periods of time and still appear profitable. As a corollary, high levels of profitability are not necessarily an indicator of good management. If a strategic position is strong enough, even fools can churn out good results (for a while).
6. Because strategic opportunities are by definition uncertain and connected to the possession of unique information or resources, strategy analysis must be situational. Just as there is no algorithm for creating wealth, strategic prescriptions that apply to broad classes of firms can only aid in avoiding mistakes, not in attaining advantage.

7. Because isolating mechanisms act to protect the first successful mover, speed is critical despite (and, in fact, because of) high levels of ambiguity. Good strategy is not necessarily enacted with a high level of initial confidence, although general management may appear confident in order to spur action. If firms wait until the proper method of entering a market or producing a product is fully understood it will normally be too late to take advantage of the information.

Conclusions

The thesis presented in this paper is that it is both possible and fruitful to attempt formal modeling of Schumpeterian, or strategic, competition. One way this can be done is to model entrepreneurs as rational maximizers with bounded knowledge, making explicit the fact that our causal models of the physical and social worlds are incomplete and frequently in error. The model of uncertain imitability is an example of this approach.

By viewing strategy as entrepreneurship that both depends upon and creates interfirm heterogeneity, I have generated a number of propositions concerning the behavior of populations of firms. I have also drawn implications for normative theory. In particular, the area in which progress would be most valuable appears to be the creation of a developmental model of industry structure.

Figure 1: Rescaling y_s

